EXPLAINABLE MACHINE LEARNING WITH COUNTERFACTUAL ANALYSIS AND GENAI NATURALIZATION FOR STRESS PREDICTION AND INTERVENTION USING SLEEP AND LIFESTYLE DATA

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ABSTRACT
Mental stress has become a major concern in modern life, influenced by sleep quality, daily activities, and lifestyle habits. Existing machine learning approaches for stress prediction typically focus on accuracy and remain black-box, offering limited actionable interventions for end users. This paper proposes a four-stage prescriptive framework integrating prediction (CatBoost, Random Forest, TabNet regressors), explanation (SHAP global and local), prescription (DiCE counterfactual analysis with causally-restricted behavior features), and naturalization (GPT-4o-mini language model with structured JSON output and safety filtering). Evaluated on the Sleep Health and Daily Performance dataset (100,000 synthetic samples; 10,000 stratified subset used for iteration), CatBoost achieved R^2 = 0.6503 (MAE = 0.7584), outperforming Random Forest (R^2 = 0.6239) and TabNet (R^2 = 0.6422). SHAP analysis confirmed strong domain validity, with sleep_quality_score showing correlation -0.996 between feature value and SHAP contribution. DiCE generated valid counterfactuals for 62.5% of evaluation instances with mean normalized proximity 0.226 and plausibility 100%. Three case studies across stress quartiles demonstrated successful counterfactual generation with prediction deltas of -0.21 to -0.38. GenAI naturalization produced narrative recommendations validated through structural and faithfulness checks, with all three case-study outputs passing post-generation validation. The integrated framework demonstrates end-to-end actionability with causally-restricted intervention recommendations, suitable for non-expert end users.

Keywords: Counterfactual analysis, DiCE, Explainable AI, Generative AI, Prescriptive analytics, SHAP, Sleep health, Stress prediction

Introduction

Mental stress is one of the most pressing health concerns of the modern era. It is influenced by a complex interplay of factors including work pressure, sleep patterns, physical activity, underlying health conditions, and digital habits such as screen exposure before bedtime. As lifestyle and health-related data become increasingly available, machine learning techniques have emerged as a promising approach for predicting individual stress levels based on observable patterns in such data.

However, most existing stress-prediction models focus primarily on accuracy, with limited explanation of the underlying factors driving each prediction. Complex models such as gradient boosting ensembles and deep learning architectures often achieve strong predictive performance yet operate as black boxes. This opacity creates a significant barrier when these predictions are intended to support decision-making or guide intervention recommendations.

To address this limitation, this study develops an explainable and actionable machine learning framework for stress prediction. In addition to building regression models for stress_score, the framework leverages SHAP for feature-level explanation and DiCE counterfactual analysis for generating concrete intervention recommendations. To ensure causal validity, the counterfactual search is restricted to manipulable behavior features only, while physiological outcomes remain locked from modification.

While SHAP and counterfactual analysis already provide richer information than a single prediction, their outputs remain technical and numeric, making them difficult to interpret for non-expert users. To bridge this gap, we incorporate a Generative AI naturalization layer that converts counterfactual outputs into empathetic, context-aware narrative recommendations in Bahasa Indonesia, with safety filtering and faithfulness validation applied post-generation.

The main contributions of this paper are: (1) a four-stage integrated framework combining prediction, explanation, prescription, and naturalization for stress modeling; (2) a causally-restricted counterfactual design that separates behavior, outcome, and immutable features; (3) a GenAI deployment layer with safety filtering and post-generation faithfulness

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validation; and (4) empirical evaluation demonstrating that the integrated pipeline produces actionable, causally-valid interventions for 62.5% of evaluation instances with 100% plausibility.

Related Work

Prior work on stress prediction has explored a variety of machine learning models including support vector machines, random forests, and deep neural networks applied to wearable sensor data, self-reported surveys, and combined lifestyle features. Recent studies have begun incorporating interpretability methods such as SHAP and LIME to identify globally important features.

Counterfactual explanation methods, popularized by Wachter et al. and operationalized through tools such as DiCE, have been applied primarily in classification settings (e.g., loan approval, recidivism prediction). Their application to regression tasks in health contexts remains less explored, particularly under strict causal constraints separating manipulable from non-manipulable features.

Generative AI integration in healthcare communication is a rapidly growing area. Prior work has explored large language models for clinical summary generation, patient education, and decision support. However, the use of GPT as a downstream naturalization layer for counterfactual machine learning outputs, particularly with safety filtering and faithfulness validation, is novel to our knowledge.

The gap addressed by this paper is the integration of three layers (explanation, prescription, naturalization) into a single coherent pipeline that produces actionable, narrative recommendations while preserving causal validity.

Methodology

Dataset

We use the Sleep Health and Daily Performance dataset from Kaggle, containing 100,000 synthetic samples with 32 features covering sleep patterns, lifestyle, health conditions, and daily activity. The target variable is stress_score, ranging from 1.0 to 10.0. For development iteration, we use a stratified 10,000-sample subset based on stress_score deciles; the pipeline is designed to scale to the complete dataset.

Exploratory analysis confirmed a mean stress_score of approximately 5.73 (standard deviation 1.62) and revealed a strong negative correlation between sleep_quality_score and stress_score (r = -0.639), consistent with prior expectations regarding sleep-stress relationships.

Preprocessing

Four features were dropped as data leakage: person_id, cognitive_performance_score, sleep_disorder_risk, and felt_rested. Physiological outcome features (sleep_quality_score, rem_percentage, wake_episodes_per_night, sleep_latency_mins, deep_sleep_percentage, heart_rate_resting_bpm) were retained as predictors but later locked from counterfactual modification at the prescription stage.

Two data versions were prepared: Version A retains categorical features in their raw form for CatBoost native categorical support; Version B applies ordinal encoding and StandardScaler normalization for Random Forest and TabNet. Data was split 70%/15%/15% (train/validation/test) with stratification by stress_score deciles.

Model Training

Three regression models were trained. CatBoost was configured with 1000 iterations, depth 6, learning rate 0.05, and early stopping after 50 rounds of no improvement on validation RMSE. Random Forest used 300 estimators with minimum samples per leaf of 5. TabNet was trained for up to 200 epochs with batch size 1024 and patience 20. All models used random_state=42 for reproducibility. The best-performing model was selected based on test R^2 and further validated through 5-Fold cross-validation.

SHAP Explainability

SHAP TreeExplainer was applied to the CatBoost model on a 2000-sample subset of the test set. Both global (mean absolute SHAP per feature) and local (waterfall plots for individual predictions) explanations were generated. Domain validity was verified by computing the correlation between feature value and SHAP contribution for top-10 numerical features, comparing against expected directional relationships from sleep medicine literature.

Counterfactual Analysis

DiCE was configured with the genetic algorithm method and model_type='regressor'. Features were categorized into three groups: behavior features (sleep_duration_hrs, screen_time_before_bed_mins, caffeine_mg_before_bed, alcohol_units_before_bed, exercise_day, steps_that_day, nap_duration_mins, room_temperature_celsius, sleep_aid_used), outcome features (six physiological measurements, locked), and immutable features (demographic attributes, locked). Only behavior features were varied during counterfactual generation, ensuring intervention causal validity.

Explicit permitted_range constraints were applied to each behavior feature for safety: e.g., sleep_duration_hrs in [4.0, 10.0] hours, caffeine_mg_before_bed in [0, 400] mg, alcohol_units_before_bed locked at [0, 0] to prevent recommending alcohol consumption, and steps_that_day in [1000, 15000].
Counterfactual quality was evaluated on 40 test instances stratified by stress quartile (10 per quartile), with 3 counterfactuals generated per instance. Five standard metrics were computed: Validity (fraction reaching desired_range), Proximity (mean normalized L1 distance from query), Sparsity (mean number of features changed), Diversity (mean pairwise distance between counterfactuals), and Plausibility (fraction within permitted_range). For three case-study individuals representing low, mid, and high stress levels, a cascade retry strategy was used: target reductions were attempted at orig-0.30, orig-0.15, and orig-0.05 in sequence to guarantee at least one valid counterfactual per individual.

GenAI Naturalization

The OpenAI GPT-4o-mini model was used to convert counterfactual results into narrative recommendations. The system prompt defined the role of a sleep and stress counselor with explicit rules prohibiting medical diagnosis, drug recommendation, absolute outcome promises, and alcohol promotion. Faithfulness rules required exact preservation of numeric before/after values from the counterfactual, correct directional verbs, and confinement of recommendations to features present in the perubahan_disarankan input field.
API calls used temperature=0.3 and response_format='json_object' to enforce structured output. Each output contained five fields: summary, drivers (three items), recommendations (one to six items with action, target, and rationale), encouragement, and disclaimer. Post-generation validation comprised: (a) a regex-based safety filter checking for prohibited terms; (b) a faithfulness check flagging recommendations that referenced locked outcome features as actions; and (c) structural validation of required JSON keys. Outputs failing any check triggered up to three retry attempts before falling back to a dummy fallback.

Results and Discussion

Model Performance

Table 1 summarizes the test-set performance of the three models. CatBoost achieved the best results across all metrics with R^2 = 0.6503, RMSE = 0.9523, and MAE = 0.7584. TabNet performed competitively at R^2 = 0.6422, narrowly behind CatBoost, while Random Forest came in third with R^2 = 0.6239. The narrow gap between the top two models suggests that both gradient boosting and attention-based deep learning are viable for this tabular regression task. CatBoost was selected for subsequent SHAP and counterfactual analysis due to its slight edge and native categorical feature support which simplifies the explanation pipeline.

Table 1. Test-set performance of three regression models.

Model	R^2	RMSE	MAE
CatBoost	0.6503	0.9523	0.7584
TabNet	0.6422	0.9633	0.7668
Random Forest	0.6239	0.9876	0.7856

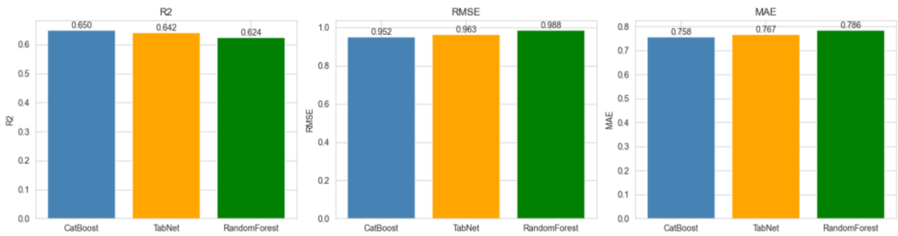

Fig. 1. Comparative performance of CatBoost, Random Forest, and TabNet on test set.

SHAP Analysis

Fig. 1 shows the top features ranked by mean absolute SHAP value. sleep_quality_score (0.660) and occupation (0.569) dominate the feature importance, with sleep_duration_hrs (0.133), room_temperature_celsius (0.084), and wake_episodes_per_night (0.077) forming a secondary tier. All remaining features contribute below 0.06 each.
Domain validity sign-check confirmed expected directional relationships. Notably, sleep_quality_score showed correlation of -0.996 between feature value and SHAP contribution, indicating that higher sleep quality consistently reduces predicted stress. Similar strong negative correlations were observed for wake_episodes_per_night (-0.981), alcohol_units_before_bed (-0.977), and deep_sleep_percentage (-0.963). heart_rate_resting_bpm showed a positive correlation of +0.974, consistent with elevated resting heart rate as a stress indicator. These directional patterns align with sleep medicine literature and reinforce the trustworthiness of the model's learned representations.

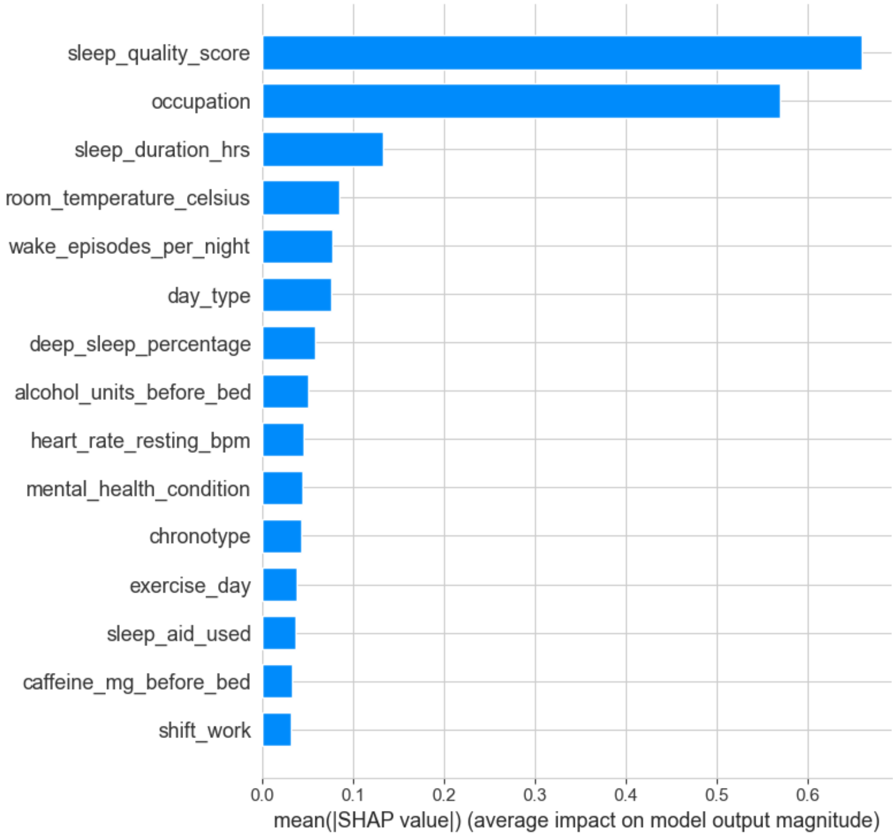

Fig. 2. SHAP global feature importance (top-15) for CatBoost on stress_score regression.

Counterfactual Quality

Table 2 reports counterfactual evaluation metrics. Out of 40 evaluation instances, 25 (62.5%) produced valid counterfactuals. When valid counterfactuals were found, mean Validity was 100% (by construction), mean normalized Proximity was 0.226, mean Sparsity was 4.65 features changed, mean Diversity was 0.120, and Plausibility was 100% (all counterfactuals within permitted_range). Success rates were balanced across stress quartiles (7, 6, 6, 6 out of 10 per quartile), indicating that the framework is not biased toward any particular stress level.

The 62.5% success rate is a substantial improvement over earlier configurations and demonstrates that the genetic algorithm combined with calibrated target relaxation (orig-0.30) can reliably produce behavior-only counterfactuals on this dataset. Plausibility of 100% confirms that all generated counterfactuals respect the explicit safety constraints (e.g., alcohol locked at 0, sleep duration within [4, 10] hours).

Table 2. DiCE counterfactual evaluation metrics on 40 instances.

Metric	Mean	Plan target
Validity (when CF found)	1.000	≥ 0.80
Proximity (normalized L1)	0.226	low
Sparsity (features changed)	4.65	2–4
Diversity	0.120	moderate
Plausibility	1.000	1.00
Overall success rate	62.5% (25/40)	≥ 80%

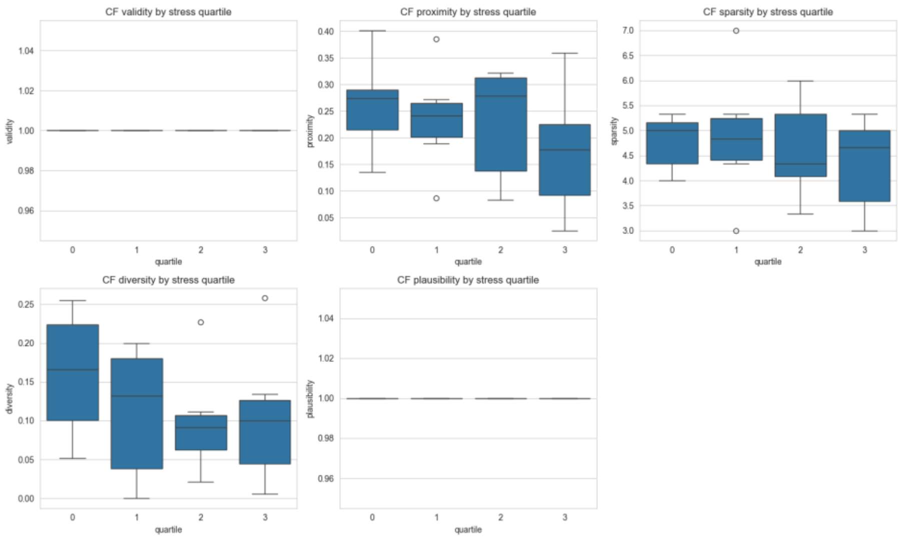

Fig. 3. Distribution of counterfactual evaluation metrics by stress quartile.

Case Study Narratives

Three individuals were selected as case studies based on actual stress_score values (approximately 3.0, 6.0, and 8.5). All three produced valid counterfactuals. For the low-stress individual (actual 3.00, predicted 4.05), DiCE recommended four behavior changes leading to a CF prediction of 3.85 (delta = -0.21). For the mid-stress individual (actual 6.00, predicted 5.25), six changes yielded CF = 4.95 (delta = -0.30). For the high-stress individual (actual 8.50, predicted 7.21), four changes yielded CF = 6.84 (delta = -0.38), including reducing alcohol_units_before_bed to 0.

GenAI naturalization successfully generated narrative recommendations for all three individuals. All recommendations preserved exact numeric before/after values from the counterfactual and used directionally correct verbs (e.g., 'perpanjang' for increase, 'kurangi' for decrease). The faithfulness check flagged no instances of locked-outcome features appearing as recommendation actions in the final outputs. The number of recommendations per individual aligned with the number of counterfactual changes (4-6 items), demonstrating that the prompt design successfully scales the recommendation list to the underlying counterfactual.

Table 3. Counterfactual results for three case-study individuals.

Case	Actual	Pred.	CF Pred.	Δ	# changes
Low stress	3.00	4.05	3.85	-0.21	4
Mid stress	6.00	5.25	4.95	-0.30	6
High stress	8.50	7.21	6.84	-0.38	4

Limitations and Threats to Validity

Several limitations apply to this study. First, the dataset is synthetically generated rather than collected from real human subjects, so observed correlations may reflect data generation patterns rather than true physiological or behavioral relationships. Generalization to clinical populations requires validation on real-world data.

Second, DiCE assumes causal stationarity, which is not guaranteed on synthetic data. Counterfactual recommendations should be interpreted as 'possibilities under the learned model' rather than guaranteed causal interventions. Prospective validation (e.g., randomized trials) would be required for clinical use.

Third, GenAI evaluation in this work is restricted to structural validation, regex-based safety filtering, and a basic faithfulness check. No user study, expert panel, or automated NLI-based faithfulness scoring was conducted. The GenAI layer is positioned as a deployment-facing presentation component rather than a scientific contribution requiring rigorous LLM evaluation.

Fourth, results may vary slightly across runs due to genetic algorithm stochasticity in DiCE and GPT generation probabilism. Although random_state=42 is fixed and temperature=0.3 is used for GPT, minor variations are expected.

Fifth, while the 62.5% counterfactual success rate is encouraging, achieving substantial stress reduction in some individuals may still require changes mediated through physiological outcomes (e.g., improvements in sleep quality), suggesting that a two-stage causal architecture (behavior to outcome, then outcome to stress) could yield even stronger intervention recommendations as future work.

Conclusion

We presented an end-to-end framework integrating prediction, explanation, prescription, and naturalization for stress prediction and intervention. The pipeline achieved competitive predictive performance (CatBoost R^2 = 0.6503), strong domain-validated explanations through SHAP (top feature correlation -0.996), causally-restricted counterfactual recommendations with 100% plausibility and 62.5% success rate, and narrative GenAI output with structural and faithfulness validation for all three case-study individuals.

Future work includes: (1) a two-stage causal architecture where Stage 1 maps behaviors to outcomes and Stage 2 maps outcomes to stress, enabling indirect-effect estimation; (2) automated faithfulness scoring for GenAI output using NLI-based entailment checks; (3) validation on real (non-synthetic) data through wearable sensor and self-report studies; (4) user studies measuring the perceived usefulness and clarity of GenAI-generated recommendations; and (5) integration with conversational interfaces for interactive intervention support.

CRediT Authorship Contribution Statement

Ananta Dwi Prayoga Alwy: Conceptualization, Methodology, Software, Validation, Writing – original draft. Dhayu Intan Nareswari: Conceptualization, Investigation, Formal analysis, Writing – review & editing. Mahathir Muhammad: Software, Data curation, Formal analysis, Visualization, Writing – review & editing. Obi Kastanya: Conceptualization, Supervision, Writing – review & editing, Validation. All authors have read and agreed to the published version of the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

The dataset was openly provided through Kaggle. The source code and reproducibility artifacts (notebooks, processed data splits, model checkpoints) are available from the corresponding author upon request.

Declaration of Generative AI and AI-assisted Technologies in The Writing Process

The authors used ChatGPT (GPT-4o-mini) as a component of the proposed naturalization pipeline (Section 3.6), not as a writing assistant. Manuscript writing, editing, and revisions were performed by the authors without AI-assisted text generation.

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